- **과적합 판단 기준**
  - **훈련 점수와 검증 점수 간 격차가 큼**
  - 훈련 점수는 높은데 검증 점수는 낮음
  - 데이터량이 증가해도 격차가 줄어들지 않음
- **전반적 성능 판단 기준**
  - 훈련 곡선은 위쪽, 검증 곡선은 아래쪽에 위치
  - **검증 점수 숫자가 높으면 좋음**
  - 훈련 점수보다는 검증 점수가 중요 (실제 성능 반영)
  - 1.0에 가까울수록 완벽한 성능
  - 0.5는 랜덤 수준 (분류 문제 기준)
- **안정성 판단 기준**
  - **검증 곡선이 매끄럽게 상승** - 학습이 잘 됨
  - 곡선이 요동치면 불안정
  - 데이터량 증가에 따라 성능 향상되는 패턴이 좋음
  - 수렴점에서 평평해지면 정상
- **일반화 성능 판단 기준**
  - **훈련-검증 점수가 비슷하게 수렴** - 이상적
  - 격차가 작을수록 좋음
  - 새로운 데이터에서도 비슷한 성능 기대 가능

1차

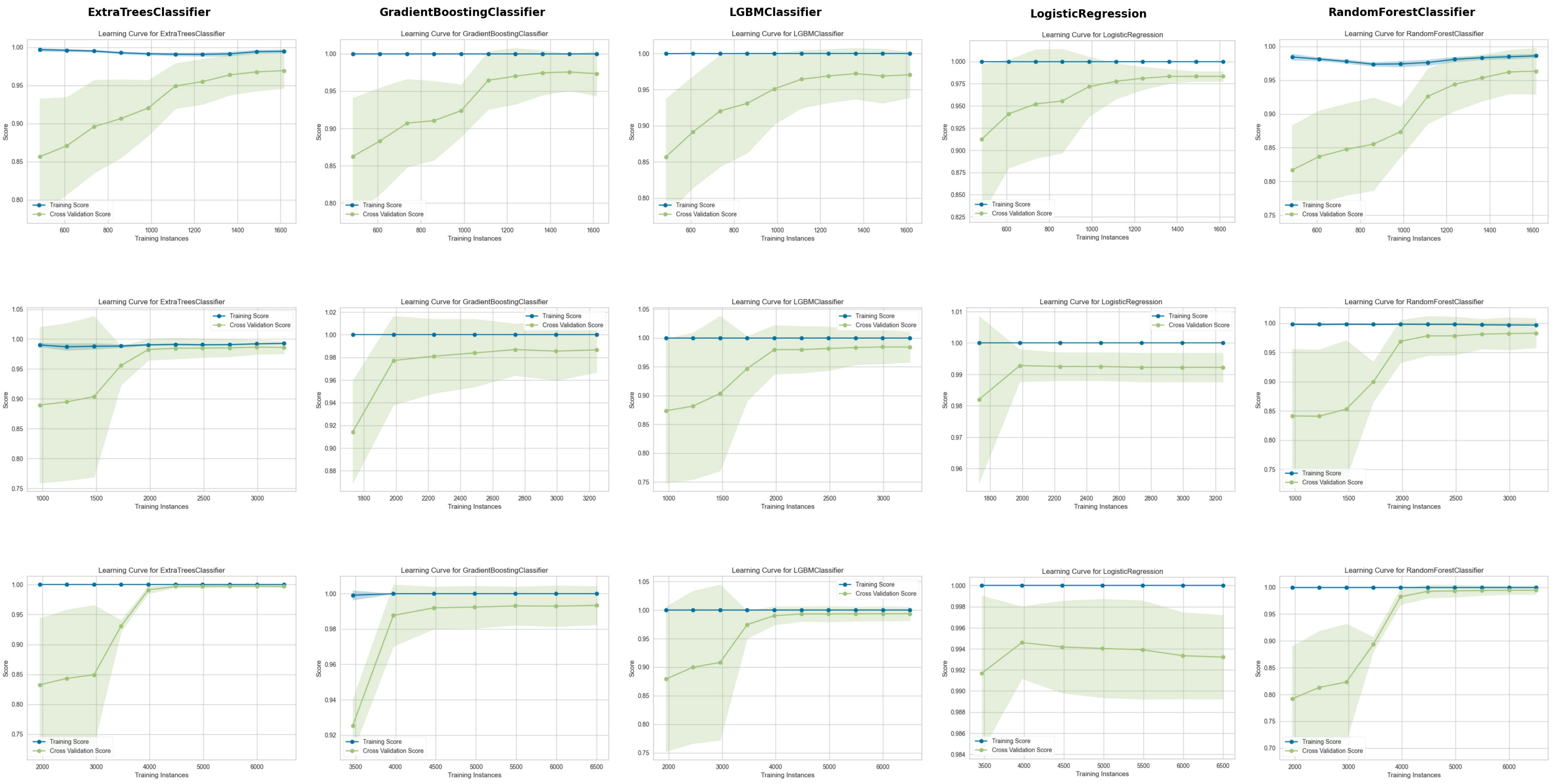
250901

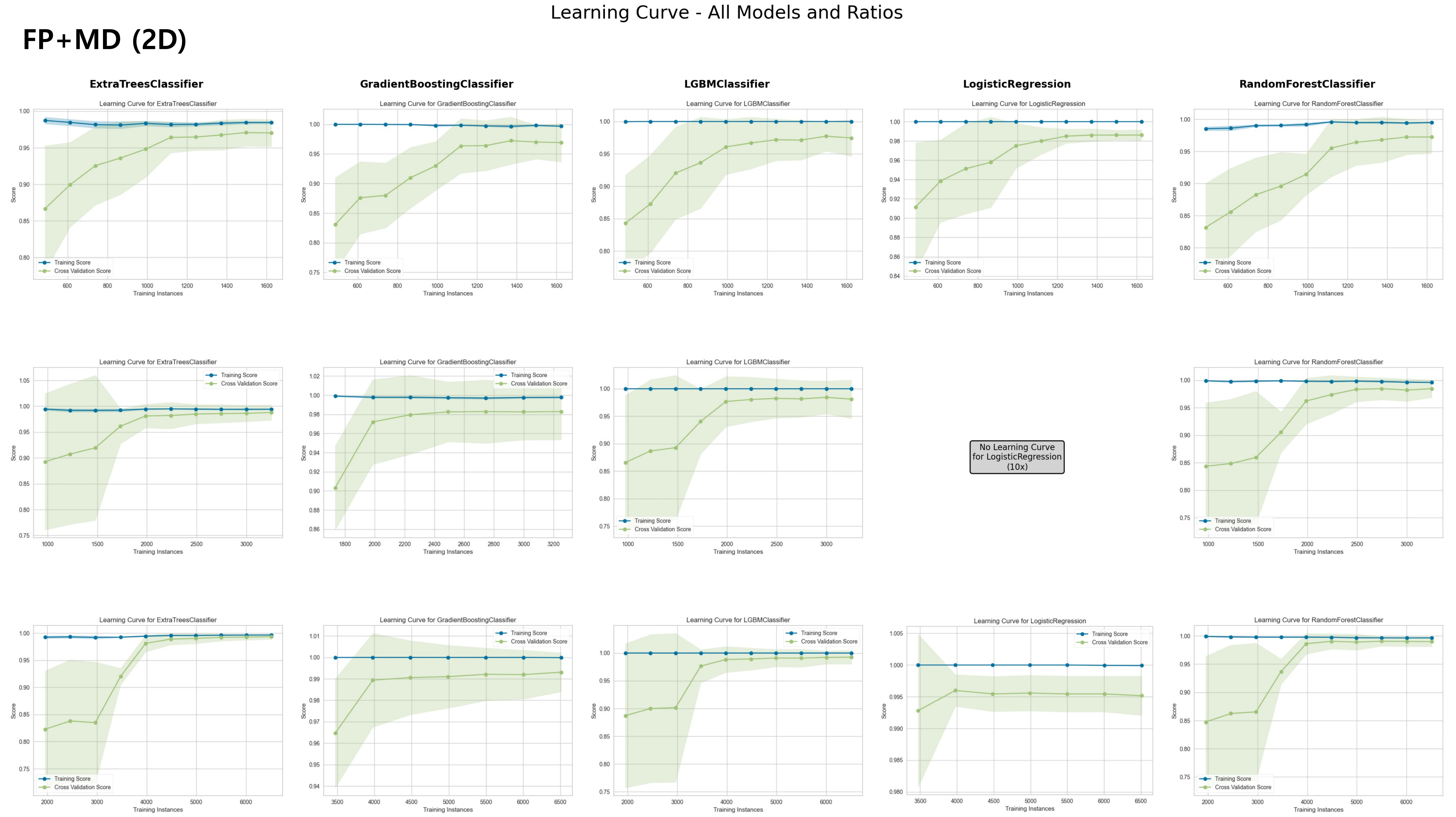
Learning Curve - All Models and Ratios

FP

Learning Curve - All Models and Ratios

FP+MD (2D+3D)

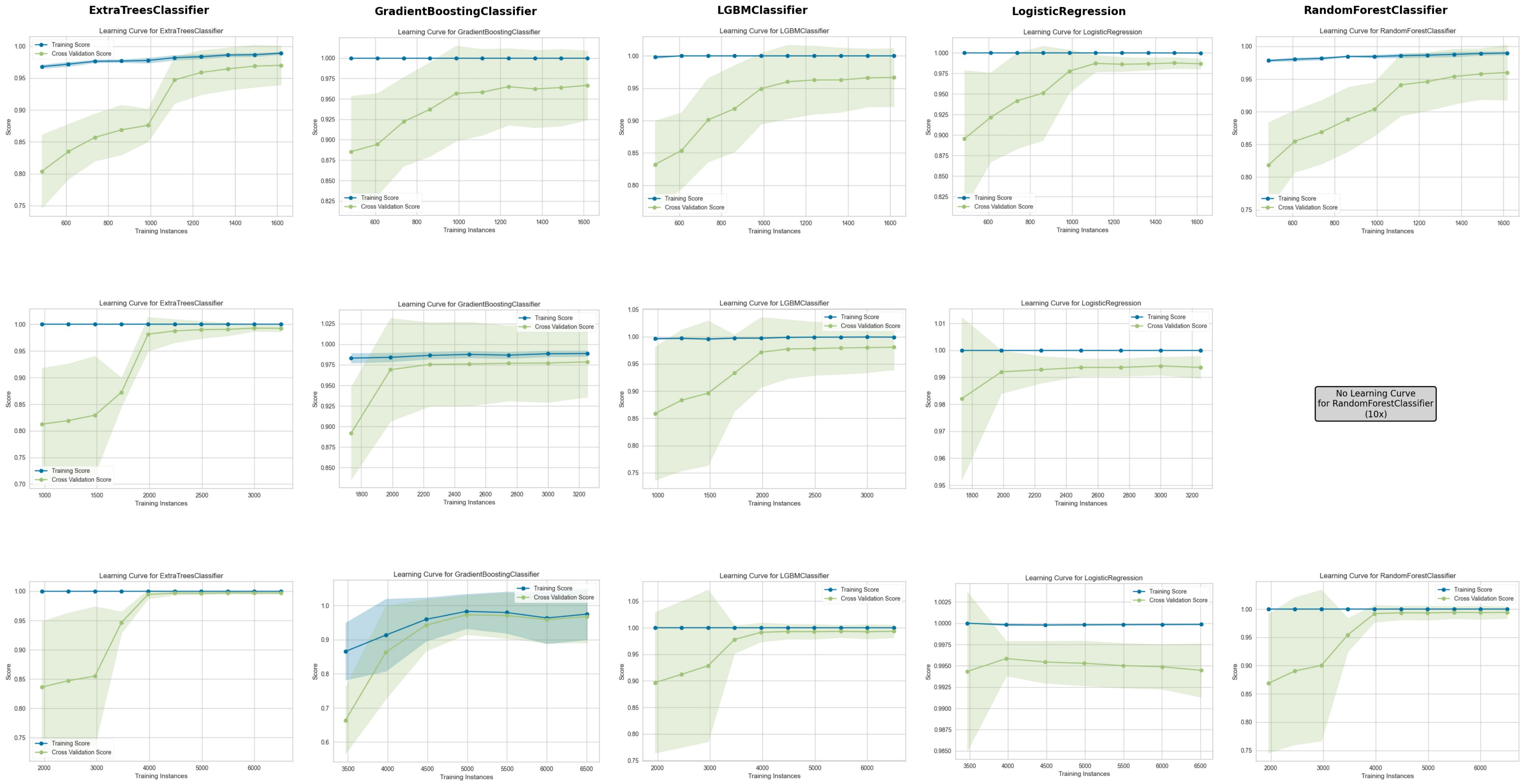




Learning Curve - All Models and Ratios
FP+MD (2D)
ExtraTreesClassifier
GradientBoostingClassifier
LGBMClassifier
LogisticRegression
RandomForestClassifier
Learning Curve for ExtraTreesClassifier
Learning Curve for GradientBoostingClassifier
Learning Curve for LGBMClassifier
Learning Curve for LogisticRegression
Learning Curve for RandomForestClassifier
Training Score
Cross Validation Score
Score
Training Instances
No Learning Curve for LogisticRegression (10x)

Learning Curve - All Models and Ratios

FP+Weka_MD (2D+3D)



```
==============================================FT0_FP+weka_MD/5x_mdT start==============================
```

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lr	Logistic Regression	0.9483	0.9643	0.8009	0.8836	0.8387	0.8080	0.8105	0.9500
et	Extra Trees Classifier	0.9483	0.9604	0.7852	0.8942	0.8358	0.8052	0.8079	0.2460
gbc	Gradient Boosting Classifier	0.9457	0.9622	0.8113	0.8586	0.8329	0.8005	0.8019	0.8940
svm	SVM - Linear Kernel	0.9448	0.9602	0.7487	0.9101	0.8194	0.7873	0.7938	0.1920
lightgbm	Light Gradient Boosting Machine	0.9439	0.9606	0.7484	0.9024	0.8172	0.7844	0.7899	0.3960
rf	Random Forest Classifier	0.9422	0.9680	0.7538	0.8842	0.8133	0.7794	0.7832	0.2500
ada	Ada Boost Classifier	0.9247	0.9202	0.7490	0.7901	0.7682	0.7233	0.7242	0.3560
dt	Decision Tree Classifier	0.8905	0.8253	0.7274	0.6616	0.6899	0.6239	0.6271	0.2180
nb	Naive Bayes	0.8888	0.7156	0.4553	0.7956	0.5784	0.5196	0.5470	0.6500
ridge	Ridge Classifier	0.7530	0.8243	0.7745	0.3850	0.5131	0.3727	0.4143	0.1960
qda	Quadratic Discriminant Analysis	0.7425	0.7651	0.7695	0.3798	0.5053	0.3609	0.4025	0.3700
lda	Linear Discriminant Analysis	0.5307	0.5507	0.6285	0.2060	0.3100	0.0774	0.1043	0.3440
knn	K Neighbors Classifier	0.2566	0.7067	0.9633	0.1794	0.3025	0.0285	0.0963	0.7720
dummy	Dummy Classifier	0.8328	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1860

```
Tuning lr...
Tuning et...
Tuning gbc...
Tuning svm...
    - [auc] Plot 생성 실패: AUC plot not available for estimators with no predict_proba attribute.
    - [calibration] Plot 생성 실패: Calibration plot not available for estimators with no predict_proba attri
Tuning lightgbm...
==============================================FT0_FP+weka_MD/5x_mdF start==============================
```

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
gbc	Gradient Boosting Classifier	0.9518	0.9620	0.8062	0.8966	0.8486	0.8200	0.8220	0.9700
lr	Logistic Regression	0.9492	0.9656	0.8063	0.8837	0.8422	0.8120	0.8140	0.2080
lightgbm	Light Gradient Boosting Machine	0.9483	0.9593	0.7695	0.9078	0.8324	0.8022	0.8063	0.4140
et	Extra Trees Classifier	0.9466	0.9673	0.7800	0.8885	0.8304	0.7988	0.8014	0.2640
rf	Random Forest Classifier	0.9422	0.9622	0.7540	0.8843	0.8135	0.7796	0.7833	0.2600
svm	SVM - Linear Kernel	0.9387	0.9545	0.7012	0.9136	0.7919	0.7568	0.7668	0.1980
ada	Ada Boost Classifier	0.9299	0.9195	0.7645	0.8075	0.7843	0.7426	0.7437	0.3620
dt	Decision Tree Classifier	0.8966	0.8373	0.7483	0.6764	0.7083	0.6459	0.6487	0.2080
nb	Naive Bayes	0.8879	0.7150	0.4553	0.7855	0.5763	0.5168	0.5425	0.2120
ridge	Ridge Classifier	0.7574	0.8303	0.7640	0.3896	0.5149	0.3763	0.4146	0.2040
knn	K Neighbors Classifier	0.2426	0.6944	0.9843	0.1792	0.3032	0.0281	0.1033	0.2040
lda	Linear Discriminant Analysis	0.5307	0.5285	0.5602	0.1923	0.2861	0.0491	0.0636	0.3500
qda	Quadratic Discriminant Analysis	0.8328	0.8318	0.0000	0.0000	0.0000	0.0000	0.0000	0.4520
dummy	Dummy Classifier	0.8328	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.1760

```
==============================================FT0_FP+weka_MD/10x_mdT start==========================
```

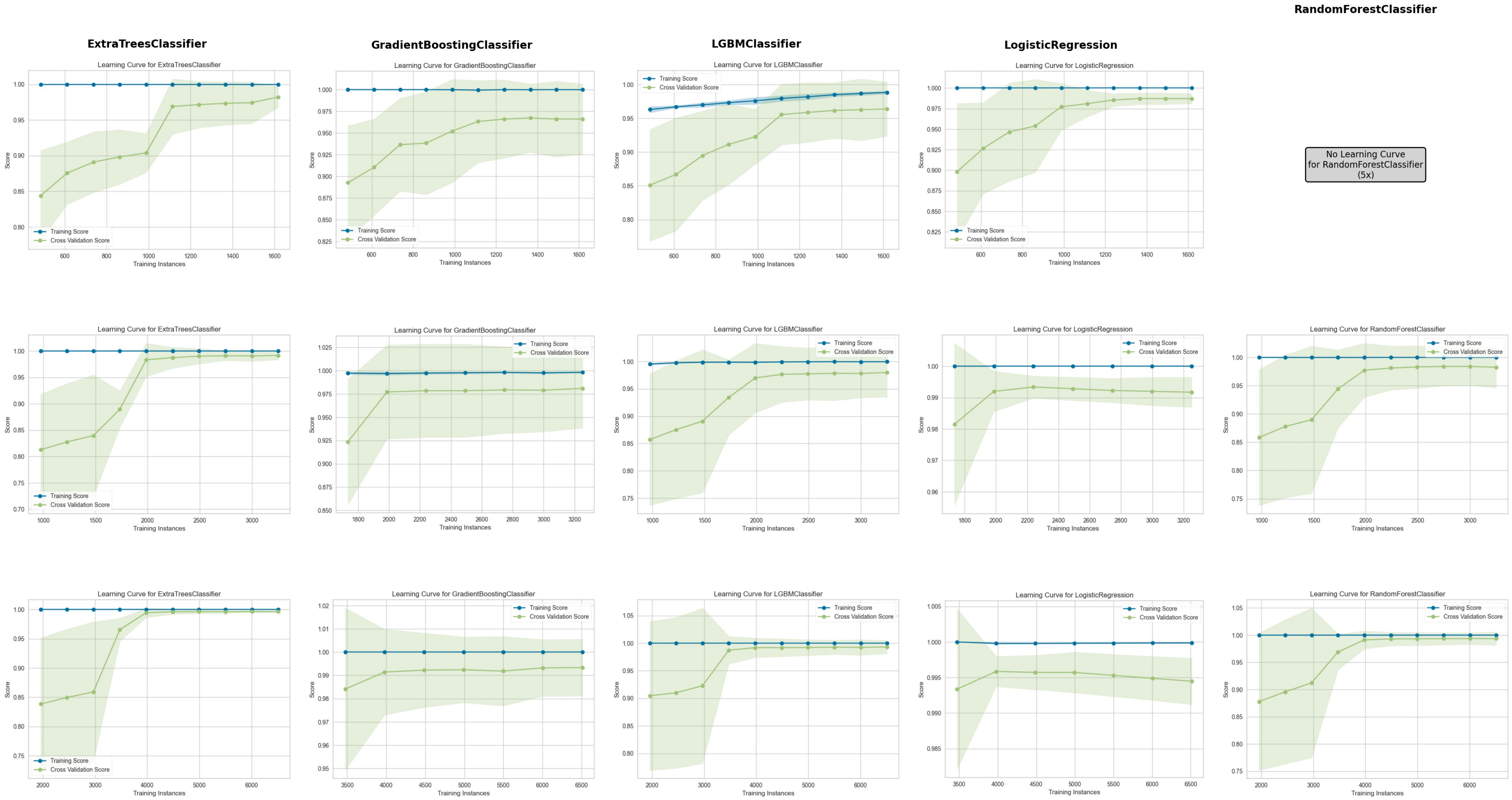
	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lr	Logistic Regression	0.9713	0.9686	0.8421	0.8430	0.8421	0.8263	0.8266	0.2860
et	Extra Trees Classifier	0.9694	0.9644	0.7737	0.8769	0.8201	0.8035	0.8065	0.3800
gbc	Gradient Boosting Classifier	0.9680	0.9681	0.7895	0.8485	0.8171	0.7996	0.8007	1.7700
lightgbm	Light Gradient Boosting Machine	0.9661	0.9647	0.7211	0.8844	0.7935	0.7753	0.7806	0.6620
rf	Random Forest Classifier	0.9642	0.9680	0.7105	0.8719	0.7817	0.7625	0.7679	0.3620
ada	Ada Boost Classifier	0.9556	0.9501	0.7526	0.7567	0.7538	0.7294	0.7299	0.5840
svm	SVM - Linear Kernel	0.9570	0.9654	0.6211	0.8745	0.7229	0.7004	0.7143	0.2520
ridge	Ridge Classifier	0.9207	0.9379	0.8421	0.5446	0.6599	0.6176	0.6373	0.2500
lda	Linear Discriminant Analysis	0.9197	0.9372	0.8421	0.5410	0.6572	0.6145	0.6347	0.4720
dt	Decision Tree Classifier	0.9326	0.8303	0.7053	0.6123	0.6551	0.6180	0.6201	0.3140
nb	Naive Bayes	0.9398	0.7300	0.4737	0.7787	0.5862	0.5560	0.5774	0.2600
knn	K Neighbors Classifier	0.1600	0.6764	0.9632	0.0947	0.1724	0.0085	0.0418	0.3020
qda	Quadratic Discriminant Analysis	0.9092	0.9088	0.0000	0.0000	0.0000	0.0000	0.0000	1.0620
dummy	Dummy Classifier	0.9092	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2420

```
Tuning lr...
Tuning et...
Tuning gbc...
Tuning lightgbm...
Tuning rf...
==============================================FT0_FP+weka_MD/10x_mdF start==========================
```

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lr	Logistic Regression	0.9713	0.9661	0.8421	0.8446	0.8426	0.8268	0.8273	0.2680
et	Extra Trees Classifier	0.9704	0.9696	0.7789	0.8811	0.8263	0.8102	0.8123	0.3520
gbc	Gradient Boosting Classifier	0.9680	0.9668	0.7895	0.8496	0.8180	0.8005	0.8014	1.7360
lightgbm	Light Gradient Boosting Machine	0.9685	0.9621	0.7579	0.8789	0.8135	0.7964	0.7993	0.6420
svm	SVM - Linear Kernel	0.9666	0.9597	0.7368	0.8797	0.7995	0.7815	0.7865	0.2440
rf	Random Forest Classifier	0.9665	0.9695	0.7158	0.8964	0.7949	0.7770	0.7834	0.3380
ada	Ada Boost Classifier	0.9589	0.9439	0.7579	0.7839	0.7699	0.7474	0.7480	0.5540
ridge	Ridge Classifier	0.9135	0.9356	0.8316	0.5181	0.6375	0.5916	0.6135	0.2520
lda	Linear Discriminant Analysis	0.9130	0.9352	0.8316	0.5159	0.6360	0.5899	0.6120	0.4600
dt	Decision Tree Classifier	0.9269	0.8248	0.7000	0.5826	0.6356	0.5954	0.5987	0.2900
nb	Naive Bayes	0.9408	0.7329	0.4789	0.7903	0.5949	0.5650	0.5865	0.2600
knn	K Neighbors Classifier	0.1567	0.6750	0.9684	0.0947	0.1726	0.0086	0.0471	0.2680
qda	Quadratic Discriminant Analysis	0.9092	0.9136	0.0000	0.0000	0.0000	0.0000	0.0000	1.0520
dummy	Dummy Classifier	0.9092	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.2220

Learning Curve - All Models and Ratios

FP+Weka_MD (2D)



2차

250902

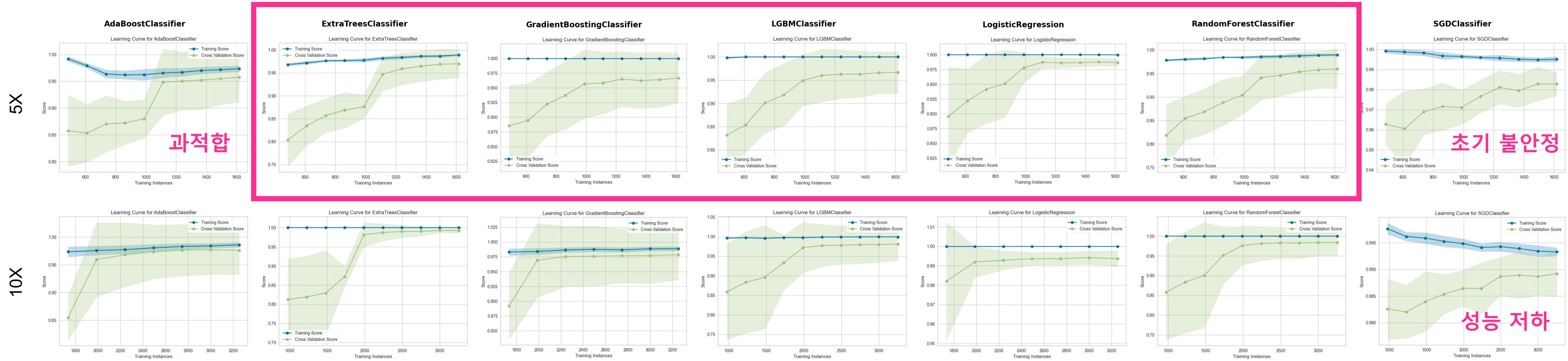
Negative data Ratio: [5x, 10x]

Model: lr, et, gbc, lightgbm, svm, rf, ada

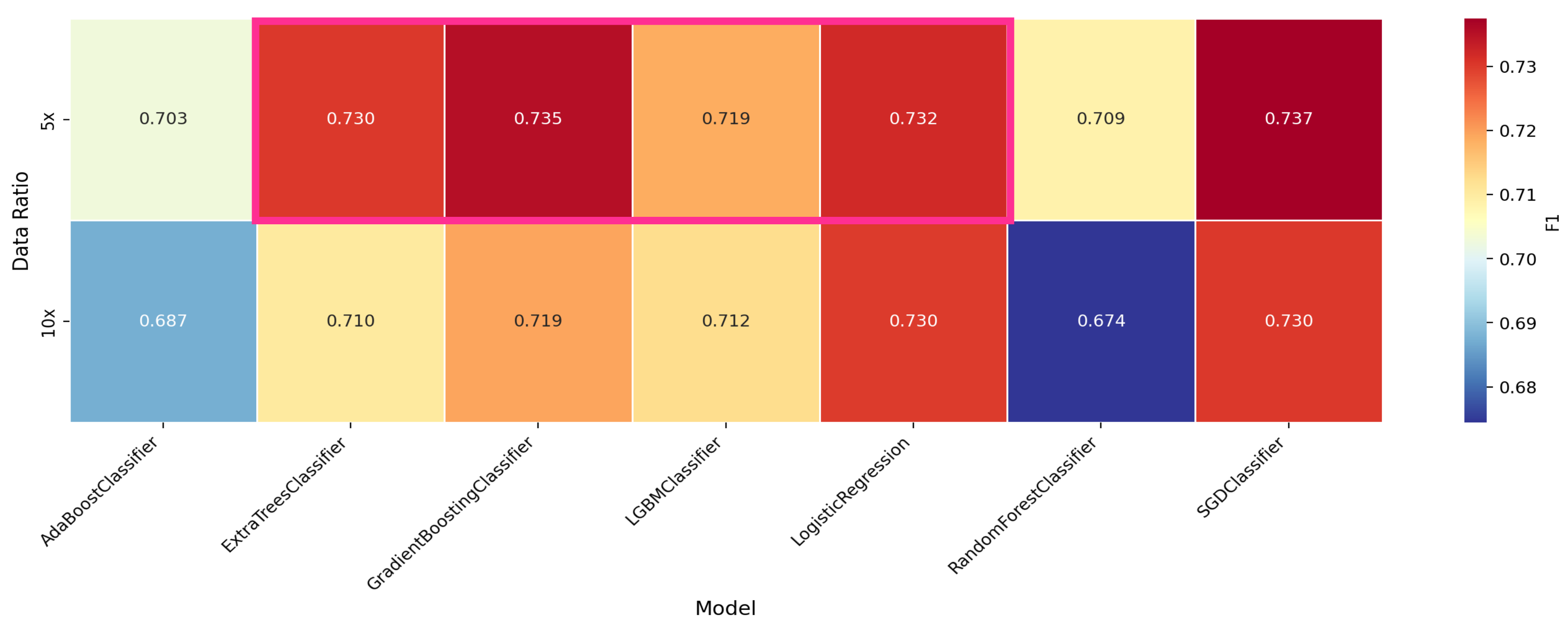
Feature: FP + Weka_MD (2D+3D)

Ratio	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
5x	LogisticRegression	0.817	0.831	0.702	0.769	0.732	0.708	0.710
	ExtraTreesClassifier	0.816	0.819	0.710	0.756	0.730	0.706	0.707
	GradientBoostingClassifier	0.818	0.827	0.709	0.768	0.735	0.712	0.713
	LGBMClassifier	0.813	0.820	0.690	0.755	0.719	0.692	0.694
	SGDClassifier	0.818	0.829	0.722	0.759	0.737	0.714	0.715
	RandomForestClassifier	0.810	0.815	0.679	0.744	0.709	0.680	0.682
	AdaBoostClassifier	0.806	0.819	0.702	0.705	0.703	0.672	0.672
10x	LogisticRegression	0.834	0.834	0.727	0.739	0.730	0.717	0.718
	ExtraTreesClassifier	0.832	0.835	0.680	0.746	0.710	0.696	0.697
	GradientBoostingClassifier	0.833	0.837	0.691	0.768	0.719	0.707	0.710
	LGBMClassifier	0.832	0.831	0.676	0.759	0.712	0.699	0.701
	SGDClassifier	0.833	0.833	0.750	0.716	0.730	0.717	0.718
	RandomForestClassifier	0.827	0.834	0.621	0.748	0.674	0.658	0.662
	AdaBoostClassifier	0.827	0.827	0.678	0.698	0.687	0.671	0.671

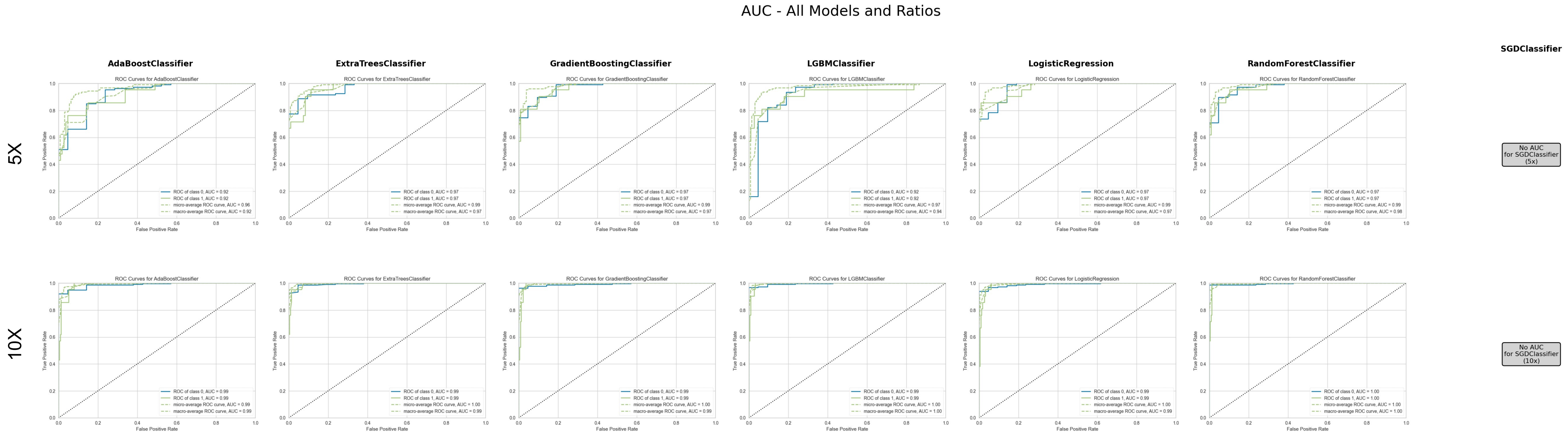
Learning Curve - All Models and Ratios

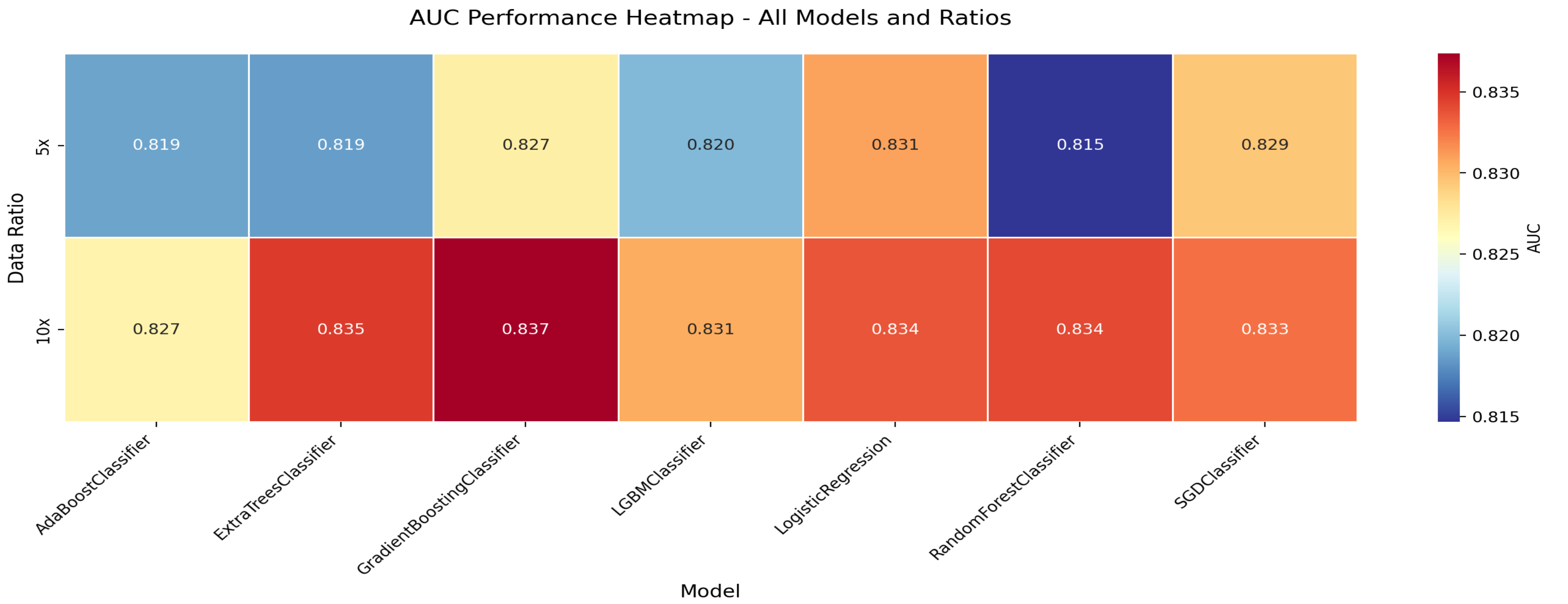


F1 Performance Heatmap - All Models and Ratios



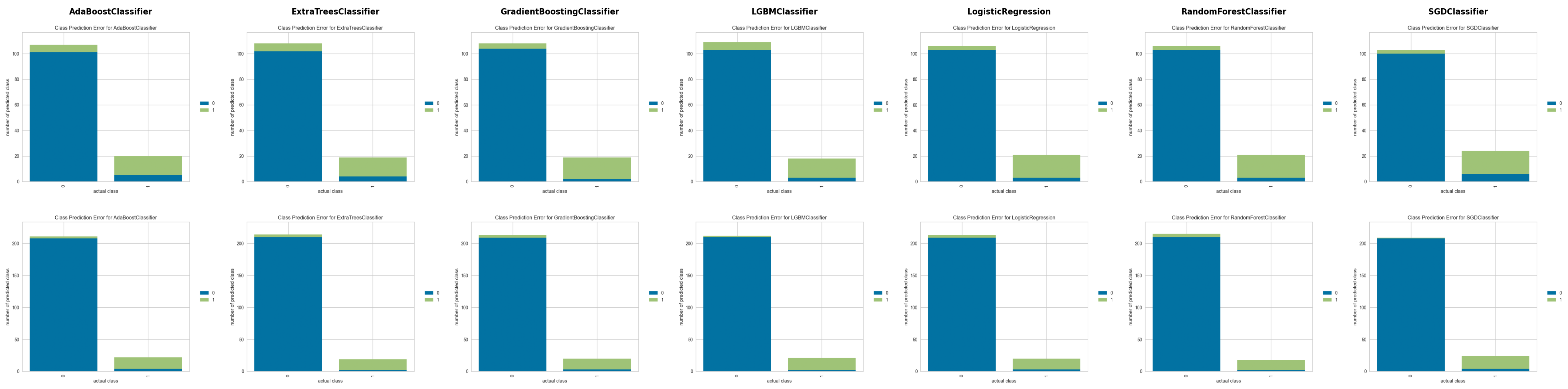
희정 의견: 5x로 'et, gbc, lgbm, lr' 네 개 모델 blend 해보는건 어때요?


AUC - All Models and Ratios
AdaBoostClassifier
ExtraTreesClassifier
GradientBoostingClassifier
LGBMClassifier
LogisticRegression
RandomForestClassifier
SGDClassifier
5X
10X
No AUC for SGDClassifier (5x)
No AUC for SGDClassifier (10x)

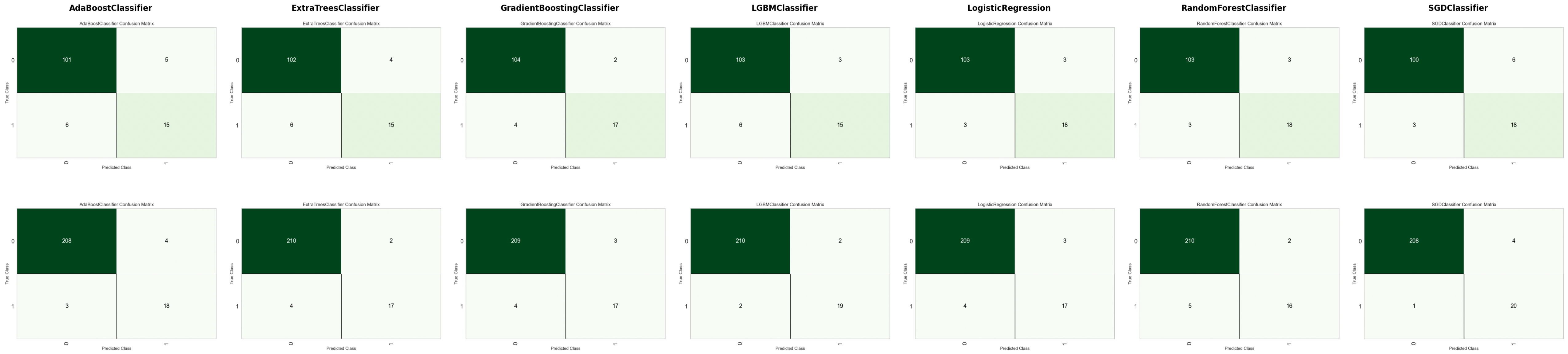

AUC Performance Heatmap - All Models and Ratios
Data Ratio
5x
10x
0.819
0.819
0.827
0.820
0.831
0.815
0.829
0.827
0.835
0.837
0.831
0.834
0.834
0.833
AdaBoostClassifier
ExtraTreesClassifier
GradientBoostingClassifier
LGBMClassifier
LogisticRegression
RandomForestClassifier
SGDClassifier
Model
AUC
0.835
0.830
0.825
0.820
0.815

Class Prediction Error - All Models and Ratios

왼쪽 바=Class 0 (파란색↑=좋음), 오른쪽 바=Class 1 (초록색↑=좋음)

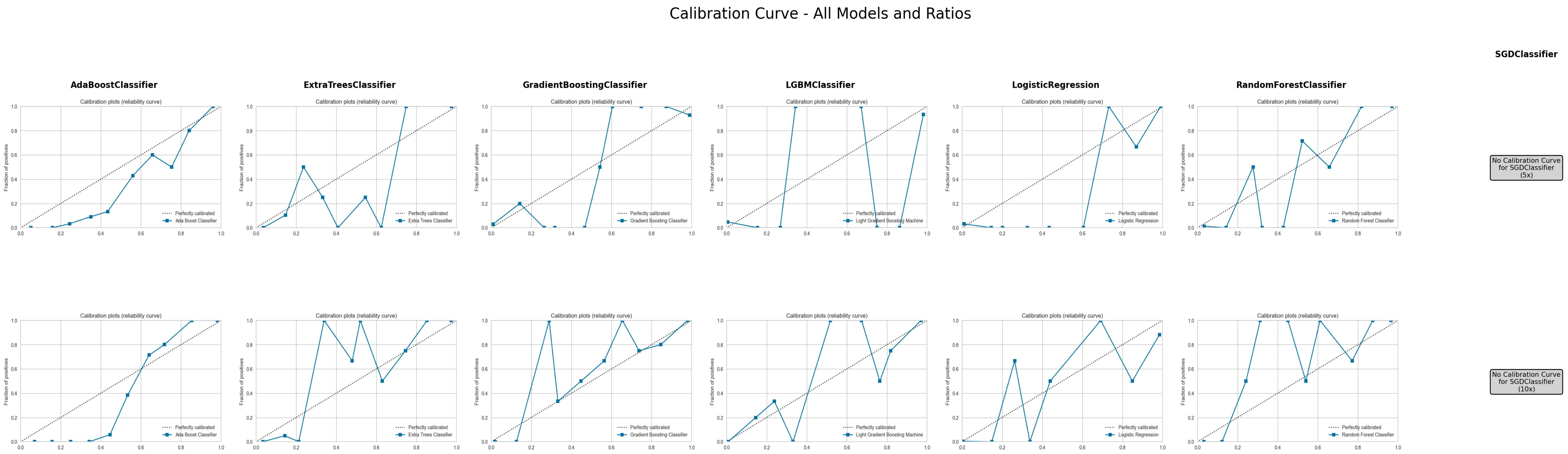


Confusion Matrix - All Models and Ratios



Calibration Curve: 예측 확률의 보정 상태, 대각선에 가까울수록 잘 보정된 모델

- X축 (Mean Predicted Probability): 모델이 예측한 확률값
- Y축 (Fraction of Positives): 실제로 양성(Class 1)인 비율



Feature Importance - All Models and Ratios